

Example: subway turnstile

Problem: Which sequences of moves will unlock turnstile?

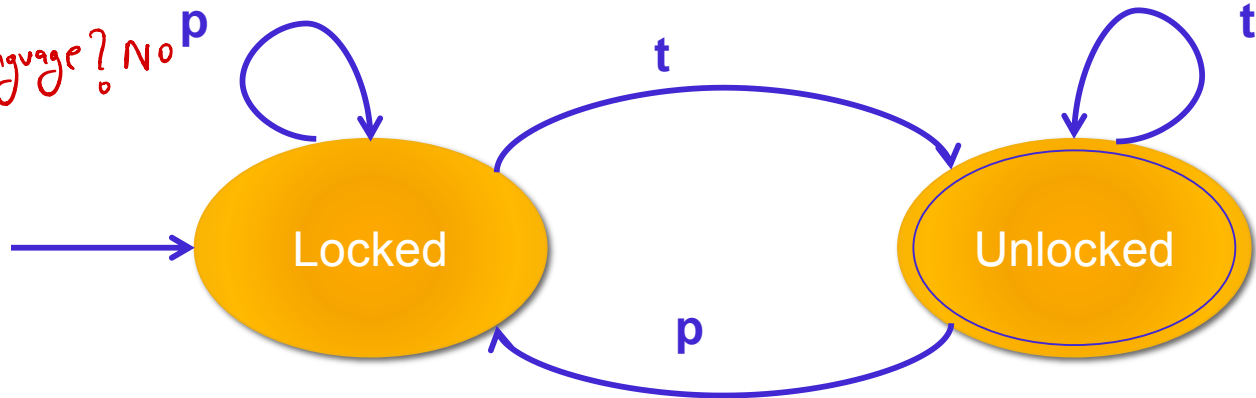
Alphabet: $\{t, p\}$

t token
p push

String: $pppt \in \text{Language? Yes}$

Language: {sequences of moves giving unlocked gate} = $\{t, ppt, \dots\}$
= {strings over alphabet that, when input to this automaton, end in Unlocked state}

$pptp \in \text{Language? No}$



What is an alphabet?

An **alphabet**, denoted by Σ , is a finite, non-empty set of symbols.

- $\Sigma = \{a, b\}$
- Unary alphabet $\Sigma = \{1\}$
- Binary alphabet $\Sigma = \{0, 1\}$
- English alphabet $\Sigma = \{a, \dots, z, A, \dots, Z\}$
- Alphanumeric alphabet $\Sigma = \{a-z, A-Z, 0-9\}$

Powers of an alphabet

$$\Sigma^* = \{\text{length } 0, \text{length } 1, \text{length } 2, \dots\}$$

$$\Sigma^* = \{\epsilon, a, b, aa, ab, ba, bb, \dots\}$$

- Σ = Some alphabet
- Σ^k = Set of all strings of length k over Σ (k can be any whole number)
- $\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$ = Set of all strings over Σ . Σ^* is the universal set of all strings
- $\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$ = Set of nonempty strings over Σ

$\{\} = \emptyset \rightarrow$ empty set

$$a^2 = a \cdot a$$

$k=0$	$k=1$	$k=2$	$k=3$
$\Sigma^0 = \{\epsilon\}$ epsilon, it's an empty string	$\Sigma = \{a, b\}$	$\Sigma^2 = \Sigma \cdot \Sigma$ = form all strings of length 2 $= \{a, b\} \cdot \{a, b\}$ $= \{aa, ab, ba, bb\}$	form all strings of length 3

What is a string?

- A **string** or word is a finite sequence of symbols chosen from Σ .
 $\varepsilon \in \Sigma^*$
- A string $\underline{x} \in \Sigma^*$. An empty string is denoted by $\underline{\varepsilon}$.
- $|x|$ = length of string x $\Sigma = \{a, b\}$ $x = abbab$ $|x| = 5$
- $n_\sigma(x)$ = #occurrences of symbol $\sigma \in \Sigma$ in the string x

$$x = abbab$$

$$n_a(abbab) = 2$$

Examples of string

- $x = \text{abaaabb}$ from $\Sigma = \{a, b\}$
- $x = 111$ from $\Sigma = \{0, 1\}$
- $x = \varepsilon$ from $\Sigma = \{a, \dots, z, A, \dots, Z\}$
- $x = \text{Bond007}$ from $\Sigma = \{a - z, A - Z, 0 - 9\}$

What is a language?

The set of strings for which our machine says yes. $\Sigma = \{a, b\}$

A language over Σ is a subset of Σ^*

$$\text{language} \subset \Sigma^*$$

- The empty language $\varnothing$.
- $\{\epsilon, a, aab\}$ - a finite language.
- ✓ Language of palindromes over $\{a, b\}$
 - $\{x \in \{a, b\}^* \mid n_a(x) > n_b(x)\}$.
 - $\{x \in \{a, b\}^* \mid |x| \geq 2 \text{ and } x \text{ begins and ends with } b\}$

$\{ab, abb, abbb, abba, \dots\}$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
No No No Yes

racecar
 $\longleftrightarrow$

Vocabulary review

Sipser p. 14

- $\{a, b, c, d, e\}$ The set whose elements are a, b, c, d, e
- $\{a, b\}^*$ The set of finite strings over a, b
 - Includes empty string ϵ
 - Includes a , aa , aaa
 - Includes b , bb , bbb
 - Includes ab , $ababab$, $aaaaaaabbbb$
 - Does **not** include infinite sequences of a 's and b 's
 - Has **infinitely many** different elements
- $|ababab| = 6$ The length of the string $ababab$ is 6
- $|\{a, b, c, d, e\}| = 5$ The size of the set $\{a, b, c, d, e\}$ is 5
- A **language** (over alphabet Σ) is a set of strings (over Σ) $L \subseteq \Sigma^*$

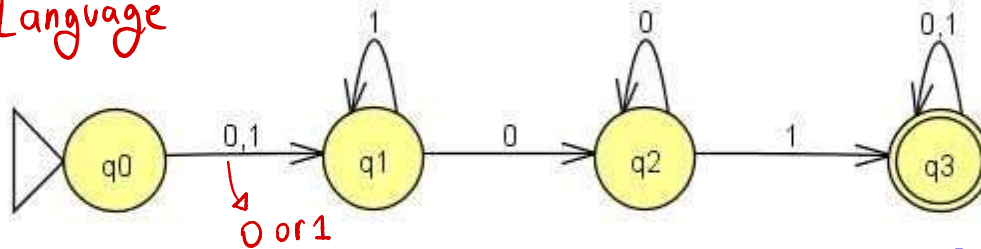
Deterministic Finite automaton

$$\Sigma = \{0, 1\}$$

Sipser p. 34, 40

- Input: finite **string** over a fixed **alphabet**
- Output: "accept" or "reject"

$x = \cancel{0}\cancel{1}\cancel{0}\cancel{1} \in \text{Language}$
 q_1



Start state
(triangle/arrow)

$x = 010 \in \text{Language? no}$

$$L = \{ x \in \Sigma^* : \text{DFA accepts } x \}$$

String $\in$ Language $\rightarrow$ Yes
(Accept state)

String $\notin$ Language $\rightarrow$ No
(Reject state)

Accept state
(double circle)

The **language** recognized by the machine is the set of strings it accepts.

Note: You need to read the Full input, and reach accept state or reach reject state.

Deterministic Finite automaton

Sipser p. 34, 40

- **Computation** of the machine on an input string

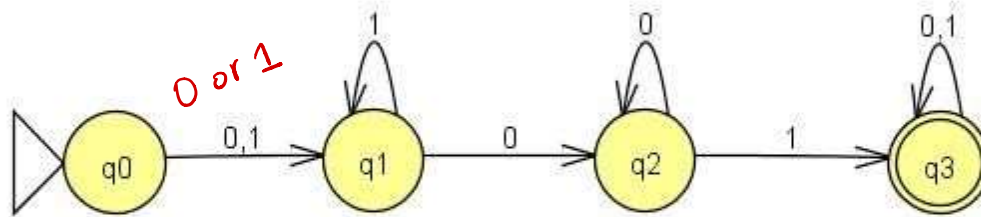
Sequence of states in the machine, starting with the initial state, determined by transitions of the machine as it reads additional input symbols.

How does the computation of the machine on a string relate to whether the string is accepted by the machine?

- A. If the **first state** in the computation is an accept state, the string is accepted.
- B. If **any state** in the computation is an accept state, the string is accepted.
- ☒ C. If the **last state** in the computation is an accept state, the string is accepted.
- D. If **all of the states** in the computation are accept states, the string is accepted.
- E. None of the above

Finite automaton

- Input: finite string over a fixed alphabet
- Output: "accept" or "reject"



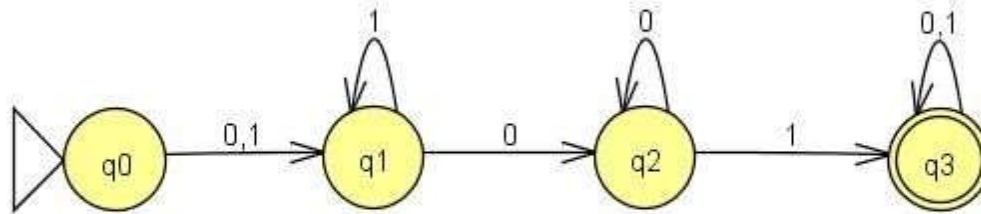
Start state
(triangle/arrow)

Accept state
(double circle)

Example input: 0001 $\in \text{Language}$

Finite automaton

- Input: finite string over a fixed alphabet
- Output: "accept" or "reject"



What is the computation of this machine on input **110**?

- A. q1, q1, q0 B. q1, q1, q2
C. q0, q1, q2, q3 D. q0, q1, q1, q2
E. None of the above

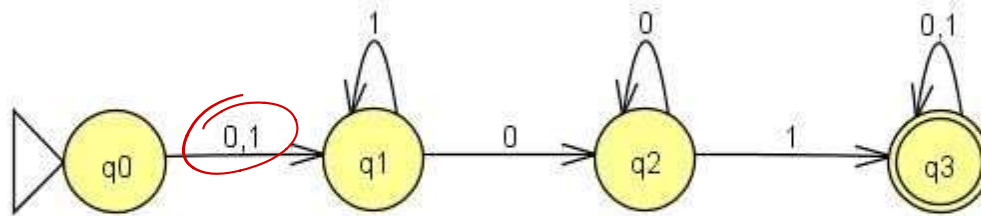
1 q_0
1 q_1
1 q_1
0 q_2

Finite automaton

- Input: finite string over a fixed alphabet
- Output: "accept" or "reject"

$\Sigma = \text{length } 0$

100 $\notin$ Language.



Is the empty string accepted? No, min length 3 accepted.

Is the string 001 accepted? Yes

Is the string 0011010101010 accepted? Yes

What is the language recognized by this machine?

All binary String which have "01" after first symbol
0 or 1

After reading full input,
if we end up in accept
state, the string gets
accepted.

For next time

- Set up course tools: Classroom
- Read all the questions + relevant examples in the book
- Read the book section (which section to read will be provided) before the class

CSE 331

AUTOMATA AND COMPUTABILITY

Lecture 2

Spring 2026

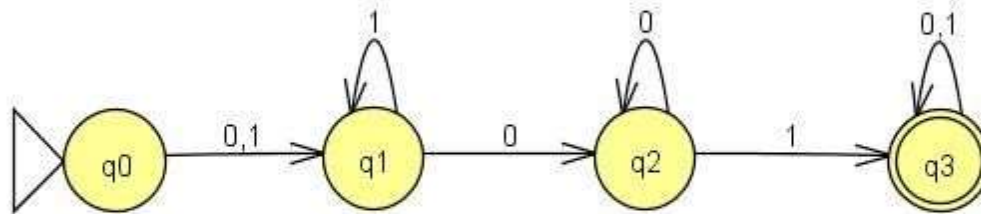
Today's learning goals

Sipser Ch 1.1

- Use and design a finite automaton via its
 - Formal definition
 - state diagram
- Identify the strings and languages accepted by a given finite automaton
- ✗ Design a finite automaton which accepts a given language
- ✗ Define the regular operations on languages

Deterministic Finite Automaton (DFA)

- Input: finite **string** over a fixed **alphabet**
- Output: "accept" or "reject"



Start state
(triangle/arrow)

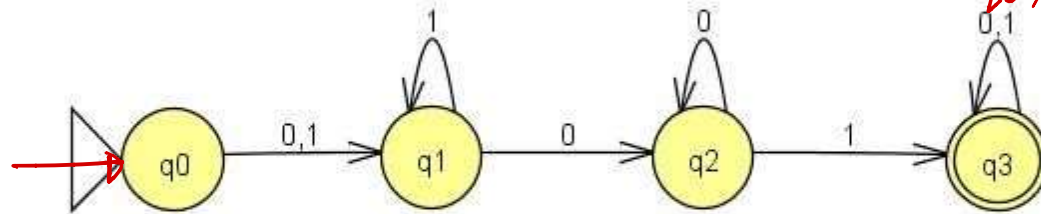
Accept state
(double circle)

Language of the machine is the set of strings it accepts

Deterministic finite automaton

- Input: finite string over a fixed alphabet
- Output: "accept" or "reject"

$$\Sigma = \{0, 1\}$$



$$Q \times \Sigma = \{(q, c) : q \in Q, c \in \Sigma\}$$

$$q_0, 0 \rightarrow q_1$$

$$q_0, 1 \rightarrow q_1$$

$$q_1, 0 \rightarrow q_2$$

$$q_1, 1 \rightarrow q_1$$

Do this for q_2 and q_3 !

Does this DFA accept the **empty string** ?

- A. Yes
- ☒ B. No
- C. I don't know

$$Q = \{ \underline{q_0}, \underline{q_1}, \underline{q_2}, \underline{q_3} \}$$

$$\Sigma = \{0, 1\}$$

q_0 = Start state

q_3 = Final state. $\subset Q$ $F = \{q_3\}$
↳ Proper subset.

Deterministic finite automaton

Sipser p. 35 Def 1.5

A **finite automaton** is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where $Q \times \Sigma = \{(q, c) : q \in Q, c \in \Sigma\}$

1. Q is a finite set called the states
2. Σ is a finite set called the alphabet
3. $\delta : Q \times \Sigma \rightarrow Q$ is the transition function
4. $q_0 \in Q$ is the start state
5. $F \subseteq Q$ is the set of accept states.

No circles and arrows, same information!

Deterministic finite automaton

Sipser p. 35 Def 1.5

A **finite automaton** is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where

1. Q is a finite set called the states
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3. $\delta : Q \times \Sigma \rightarrow Q$ is the transition function
4. q_0 $\in Q$ is the start state
5. $F \subseteq Q$ is the set of accept states.

Can there be more than one **start state** in a finite automaton?

- A. Yes, because of line 4.
- ☒ B. No, because of line 4.
- C. I don't know

Deterministic finite automaton

Sipser p. 35 Def 1.5

A **finite automaton** is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where

1. Q is a finite set called the states
2. Σ is a finite set called the alphabet
3. $\delta : Q \times \Sigma \rightarrow Q$ is the transition function
4. $q_0 \in Q$ is the start state
5. $F \subseteq Q$ is the set of accept states.

$$\Sigma = \{0, 1\}$$

$$\Sigma = \{0, 1, 2\}$$

How many outgoing arrows from each state?

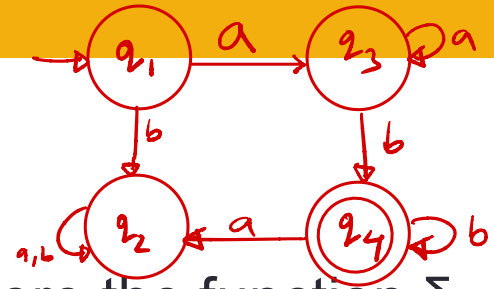
- A. May be different number at each state.
- B. Must be 2.
- C. Must be $|Q|$.
- ☒ D. Must be $|\Sigma|$
- E. I don't know.

An example

Define $M = (\{q_1, q_2, q_3, q_4\}, \{a, b\}, \delta, q_1, \{q_4\})$ where the function δ is specified by its table of values:

Input in $Q \times \Sigma$	Output in Q
(q_1, a)	q_3
$(\underline{q_2}, a)$	q_2
(q_3, a)	q_3
$(q_4, \underline{a})$	q_2

Input in $Q \times \Sigma$	Output in Q
$(q_1, \underline{b})$	<u>q_2</u>
(q_2, b)	q_2
(q_3, b)	q_4
(q_4, b)	q_4



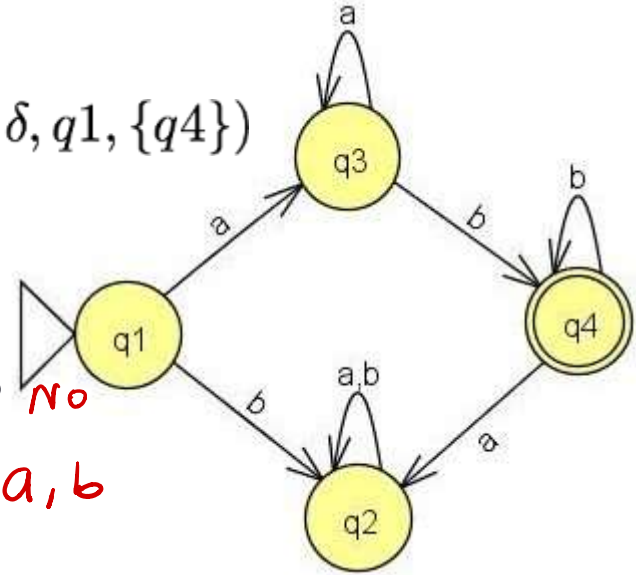
Draw the state diagram for the DFA with this formal definition.

An example

$(\{q1, q2, q3, q4\}, \{a, b\}, \delta, q1, \{q4\})$

What's an example of a

- length 1 string accepted by this DFA? *No*
- length 1 string rejected by this DFA? *a, b*
- length 2 string accepted by this DFA? *ab*
- length 2 string rejected by this DFA? *All the rest*

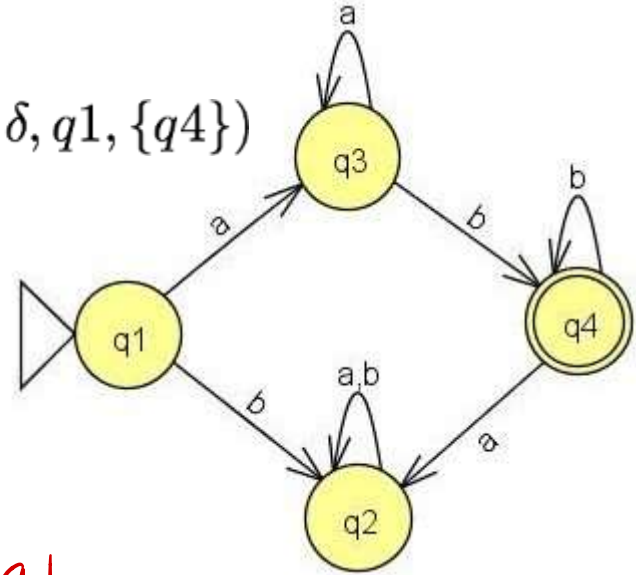


An example

$(\{q1, q2, q3, q4\}, \{a, b\}, \delta, q1, \{q4\})$

What's the best description of the language recognized by this DFA?

- A. Starts with b and ends with a or b
- B. Starts with a and ends with a or b
- ☒ C. a's followed by b's
- D. More than one of the above
- E. I don't know.



ab
a a b
a b b b

and using set notation?

An example

$(\{q1, q2, q3, q4\}, \{a, b\}, \delta, q1, \{q4\})$

This DFA recognizes
the language of all strings
of the form a's followed by b's

i.e. $\{a^n b^k \mid \underline{n}, \underline{k} \geq 1\}$

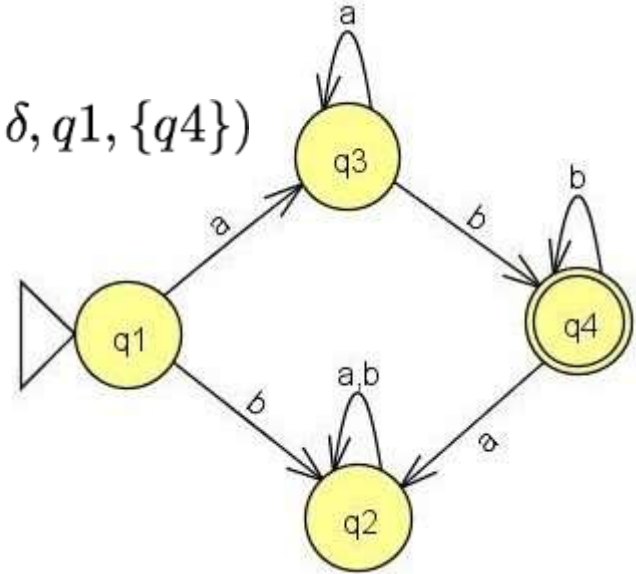
$\left. \begin{matrix} n \\ k \end{matrix} \right\}$ any value

$n = 3$

$k = 5$

$a^3 b^5 - a a a b b b b b$

$a^n = \underbrace{a a \dots a}_{n \text{ times}}$



Specifying an automaton

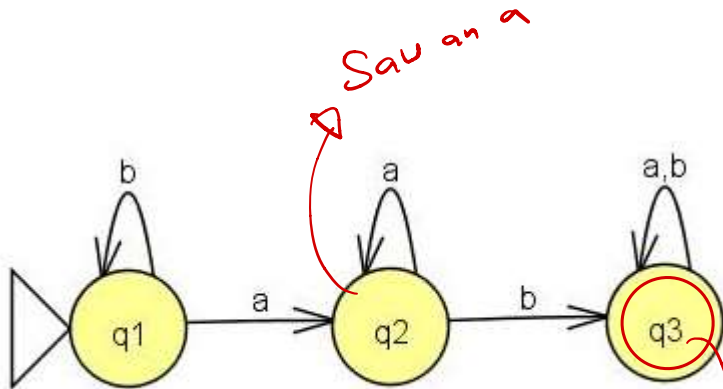
({q1,q2,q3}, {a,b}, δ , q1, ?)

$$\Sigma = \{a, b\}$$

What state(s) should be in F so that the language of this machine is
 $\{ w \mid \underline{a}b \text{ is a substring of } w \}$

- A. {q2}
- ☒ B. {q3}
- C. {q1,q2}
- D. {q1,q3}
- E. I don't know.

$$\underline{w} \in \Sigma^*$$



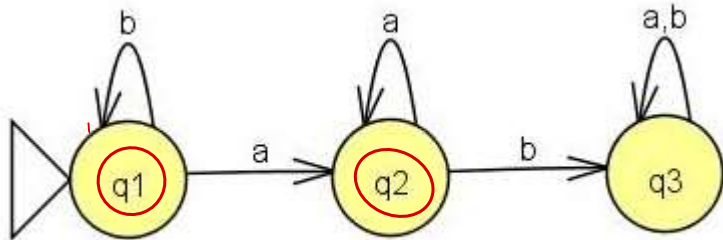
$bba\underline{a}bba \in \text{Language}$

Saw a b

Substring = Consecutive chars

Specifying an automaton

($\{q_1, q_2, q_3\}$, $\{a, b\}$, δ , q_1 , ?)



What state(s) should be in F so that the language of this machine is $\{w \mid \text{b's never occur after a's in } w\}$?

- A. $\{q_2\}$
- B. $\{q_3\}$
- C. $\{q_1, q_2\}$
- D. $\{q_1, q_3\}$
- E. I don't know.

$\begin{array}{c} b a a \\ b b a \\ b b b b \checkmark \\ a \end{array} \right\} \checkmark \text{ Yes}$

$b a \underline{a} \underline{b} \times \text{ No}$